

Math Primer

(3. Complex Numbers)

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Euler
(1707-1783)

Polynomials and Roots

- ▶ A 'n' degree polynomial is represented as

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$$

- ▶ The highest power of x is the degree of the polynomial
- ▶ e.g., $f(x) = ax + b$ (degree 1 or 'linear') , $f(x) = ax^2 + bx + c$ (degree 2 or 'quadratic')
- ▶ The number of solutions to $f(x) = 0$ are called the roots of the polynomial
- ▶ An 'n' degree polynomial will have 'n' roots , e.g.
 - ▶ A linear polynomial has 1 root $x = -b/a$,
 - ▶ A quadratic polynomial has 2 roots $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

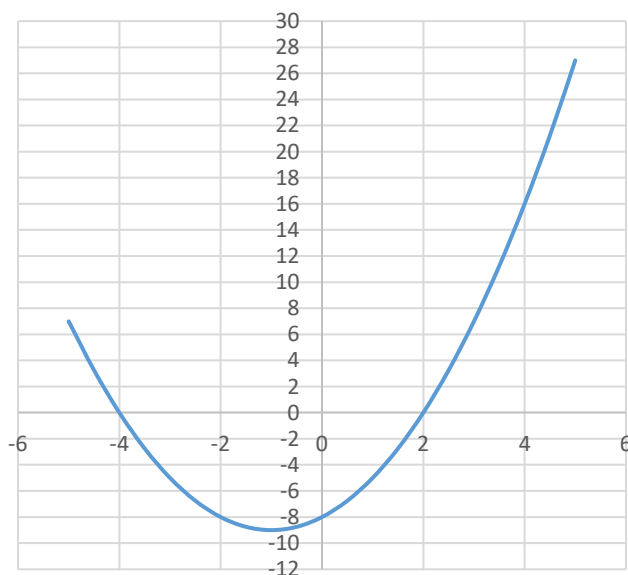
Special Case of Quadratic Function

▶ A quadratic function has 2 roots

- ▶ If $D = (b^2 - 4ac) > 0$, then it has two 'real roots' (function crosses X axis at 2 distinct points)
- ▶ If $D = (b^2 - 4ac) = 0$, then it has one repeated real root (function touches X axis at a point)
- ▶ If $D = (b^2 - 4ac) < 0$, then it has no real roots (function never crosses X axis)

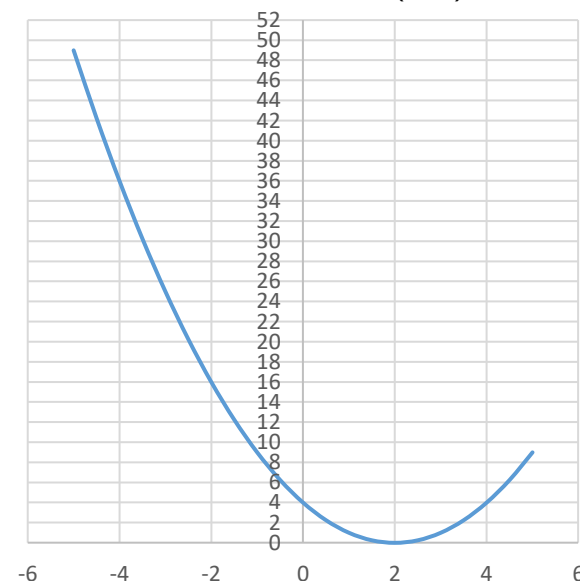
$$f(x) = x^2 + 2x - 8$$

$D = 36$, roots = (2,-4)



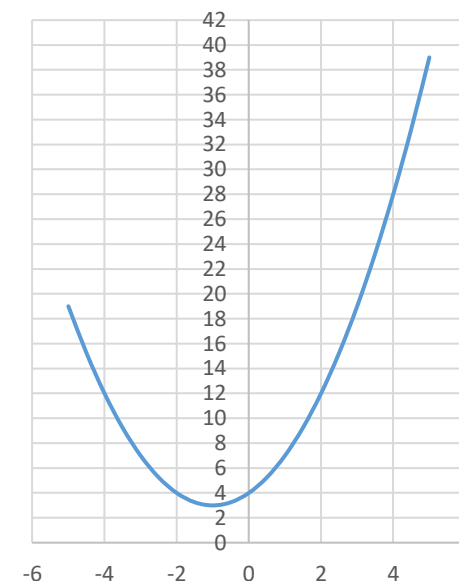
$$f(x) = x^2 - 4x + 4$$

$D = 0$, roots = (2,2)



$$f(x) = x^2 + 2x + 4$$

$D = -12$, roots = ? ?



Complex Numbers



- ▶ Let's call $i = \sqrt{-1}$ as the imaginary number (Some books adopt the notation j)
- ▶ Note the following properties : $i^2 = -1$, $i^3 = -i$, $i^4 = 1$
- ▶ A complex number is written as $(a + ib)$, where a represents the real part and b the imaginary part
- ▶ e.g., $(2 + 3i)$, $\text{Re}(2+3i) = 2$, $\text{Im}(2+3i) = 3$
- ▶ Then polynomials not having real roots can be seen as having complex roots
- ▶ e.g., $x^2 + 1 = 0$ has roots $(i, -i)$ and $x^2 + 2x + 2$ has roots $(-1 + i, -1 - i)$
- ▶ If $z = a + ib$, then conjugate of z is denoted as $\bar{z} = a - ib$
- ▶ Complex roots occur in conjugate pairs

Algebra of Complex Numbers



► Addition / Subtraction :

$$z_1 = a + ib, z_2 = c + id, \text{ then } z_1 + z_2 = (a + c) + i(b + d), z_1 - z_2 = (a - c) + i(b - d)$$

$$\text{e.g., } z_1 = 6 + 3i, z_2 = 2 - i, z_1 + z_2 = 8 + 2i, z_1 - z_2 = 4 + 4i$$

► Multiplication :

$$z_1 = a + ib, z_2 = c + id, \text{ then } z_1 z_2 = (ac - bd) + i(ad + bc)$$

$$\text{e.g., } z_1 = 3 + 5i, z_2 = 2 + 6i, z_1 z_2 = -24 + 28i$$

► Inverse :

$$z = a + ib, z^{-1} = \frac{1}{z} = \frac{a}{a^2 + b^2} + i \frac{-b}{a^2 + b^2}$$

$$\text{e.g., } z = 2 - i, z^{-1} = \frac{2+i}{5}$$

► Division :

$$z_1 = 6 + 3i, z_2 = 2 - i, \text{ then } \frac{z_1}{z_2} = (6 + 3i) \left(\frac{1}{2-i} \right) = (6 + 3i) \left(\frac{2+i}{5} \right) = \frac{(9+12i)}{5}$$

Complex Plane



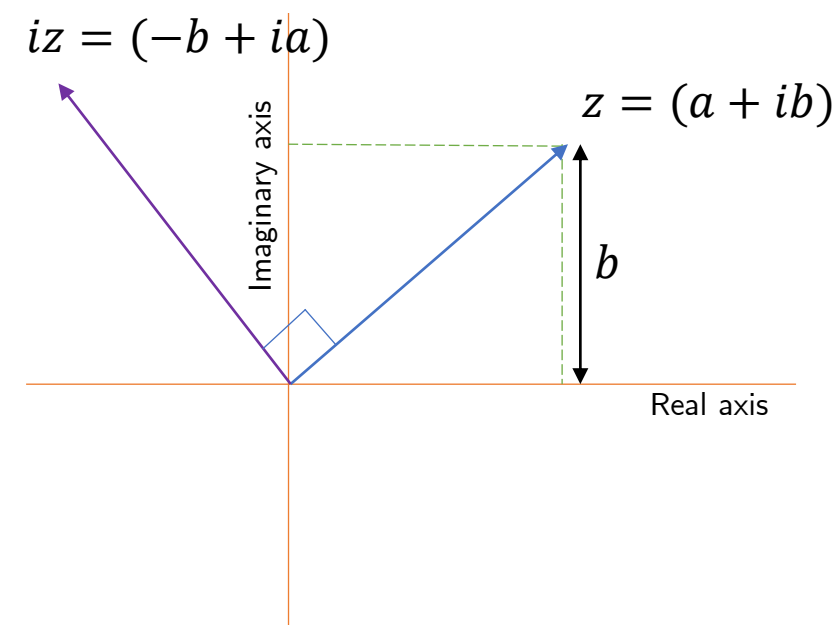
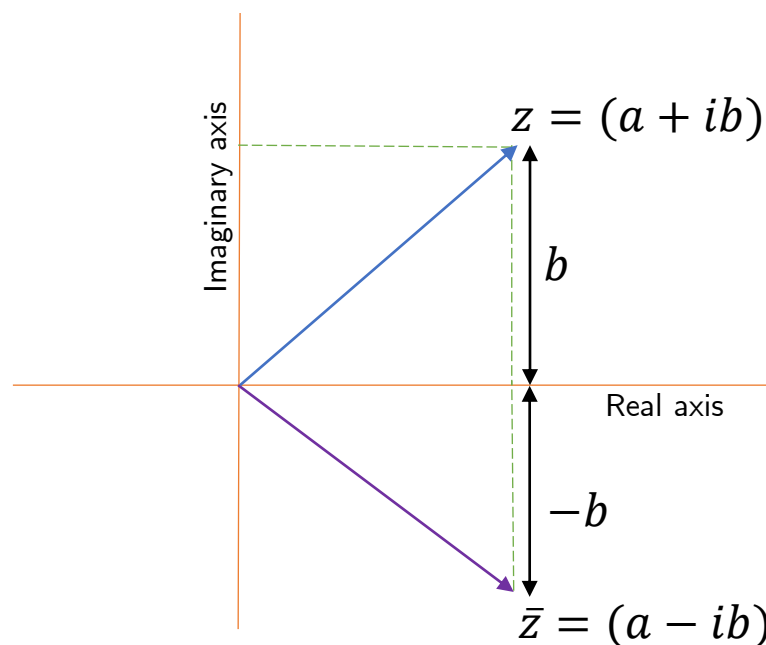
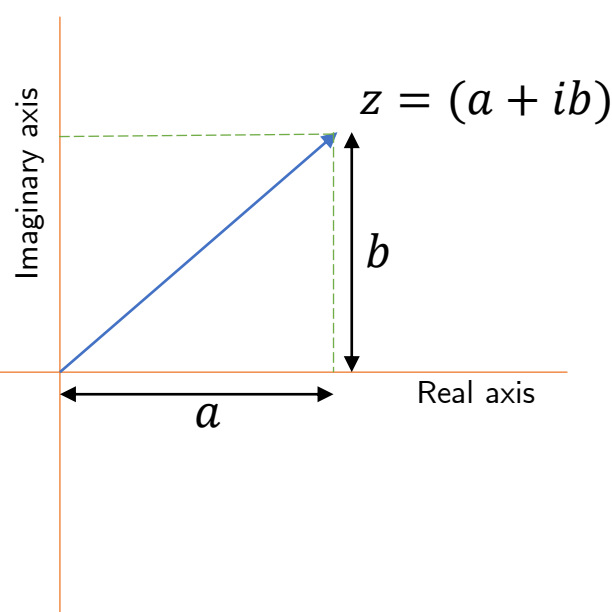
► Modulus :

$z = a + ib$, $|z| = \sqrt{a^2 + b^2}$, called the modulus which is the length of the vector

Conjugate $z = a - ib$ is the mirror image on z on X-axis

Also, $z\bar{z} = |z|^2$

Multiplying i with z , rotates it by 90° anti-clockwise



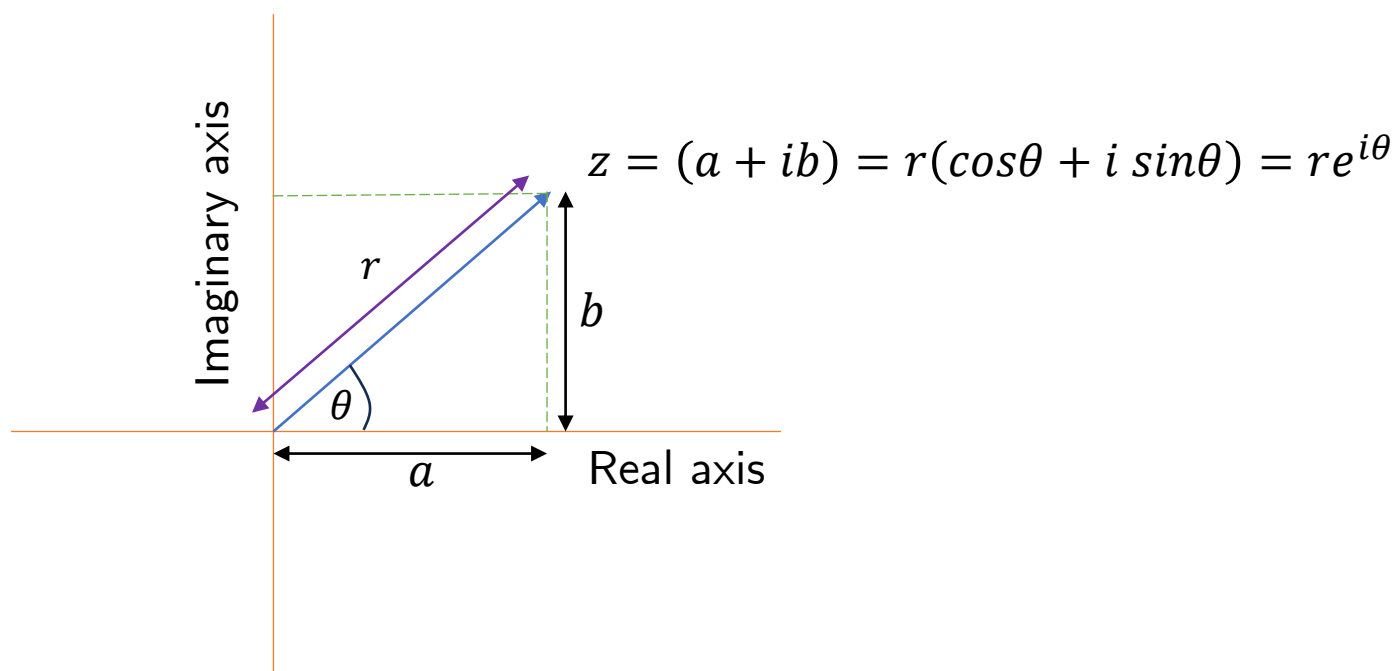
Polar Representation



- A complex number can be represented with the length and the angle it makes with X axis :

$z = a + ib = r(\cos\theta + i \sin\theta) = re^{i\theta}$, where $r = |z| = \sqrt{a^2 + b^2}$, θ is the angle with x-axis which is called as the argument of z given as $\theta = \tan^{-1}\left(\frac{b}{a}\right)$

$e^{i\theta} = \cos\theta + i \sin\theta$ is called as Euler Identity



Important Results

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

$$e^{i\pi} + 1 = 0$$

Euler's Identity

$$(\cos \theta + i \sin \theta)^n = e^{in\theta} = \cos (n\theta) + i \sin (n\theta)$$

De Moivre's Theorem

$$e^{\pm z} = 1 \pm z + \frac{1}{2!}z^2 \pm \frac{1}{3!}z^3 + \dots$$

$$\sin z = z - \frac{1}{3!}z^3 + \frac{1}{5!}z^5 - \dots$$

$$\cos z = 1 - \frac{1}{2!}z^2 + \frac{1}{4!}z^4 - \dots$$

Power Series

Exercises



1. If $(x + iy)^3 = u + iv$, then show that $\frac{u}{x} + \frac{v}{y} = 4(x^2 - y^2)$
2. If $\left(\frac{1+i}{1-i}\right)^m = 1$, then find the least m
3. Find $z \in \mathbb{C}$ such that $z^2 \in \mathbb{R}$
4. Prove that there is no complex number z , such that $|z| - z = i$
5. Express $\ln(a + ib)$ in standard form